

Adaptive Contrastive Decoding in Retrieval-Augmented Generation for Handling Noisy Contexts

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Abstract

When using large language models (LLMs) in knowledge-intensive tasks, such as open-domain question answering, external context can bridge a gap between external knowledge and LLM’s parametric knowledge. Recent research has been developed to amplify contextual knowledge over the parametric knowledge of LLM with contrastive decoding approaches. While these approaches could yield truthful responses when relevant context is provided, they are prone to vulnerabilities when faced with noisy contexts. We extend the scope of previous studies to encompass noisy contexts and propose adaptive contrastive decoding (ACD) to leverage contextual influence effectively. ACD demonstrates improvements in open-domain question answering tasks compared to baselines, especially in robustness by remaining undistracted by noisy contexts in retrieval-augmented generation.

1 Introduction

While large language models (LLMs) (Touvron et al., 2023; Achiam et al., 2023) achieve remarkable performance levels across diverse benchmarks, they sometimes struggle to generalize to knowledge-intensive tasks such as open-domain question-answering (QA; Chen et al., 2017) and may also fail to capture long-tail knowledge, leading to unfaithful output generation (Mallen et al., 2023; Kandpal et al., 2023). One common approach to address these issues is fine-tuning the model, but this results in a quadratic rise in computational demands as the size of the LLMs increases exponentially (Longpre et al., 2023). To address these issues, researchers have been investigating strategies to combine non-parametric knowledge with LLMs during response generation without explicit re-training. This approach leverages external information from knowledge bases and enhances the capability of the LLMs dynamically, ensuring that the information is both current and accurate.

Early studies in this field attempt to append relevant context information to generate more accurate responses (Asai et al., 2023a). Especially, contrastive decoding (Li et al., 2023; Malkin et al., 2022; Liu et al., 2021) yields significant enhancement in various tasks by amplifying the influence of the given context at decoding step (Shi et al., 2023; Zhao et al., 2024). While such methods work well when context information is correct and faithful, in real-world scenarios, context information is not always correct and may contain some noisy and unfaithful information. For instance, if the retrieval system pulls in irrelevant or contradictory information, it could lead to incorrect responses (Wang et al., 2024; Wu et al., 2024; Yu et al., 2024). This highlights the necessity for a generation model that can gauge the appropriateness of the context by itself, being robust to noise and unfaithful data to ensure the output remains reliable (Yoran et al., 2024).

To assess whether the existing contrastive decoding approaches can be utilized in practice, we extend the setting to where the gold-standard context is not guaranteed, specifically in the retrieval-augmented generation (RAG) framework (Yao et al., 2022; Shi et al., 2024; Izacard et al., 2023). In this paper, we demonstrate that existing context-aware contrastive decoding approaches experience performance drops in open-domain question answering, especially when the retrieved context is noisy. To address this issue, we propose **adaptive contrastive decoding (ACD)**, adaptively weighting the contrastive contextual influence on the parametric knowledge, making it suitable for noisy context settings.

Incorporating the distinction between contextual and parametric knowledge, our approach aims to mitigate the dominance of potentially noisy contextual information in model output. We control contrastive contextual influence based on context’s contribution to LLM’s uncertainty reduction, thereby

minimizing its disruptive effect during decoding. Through in-depth experiments with three open-domain QA datasets, we demonstrate the potential of the proposed approach with increased overall performance. Moreover, ACD enhances the performance significantly on the noisy context scenario while minimizing performance degradation on the gold context scenario compared to the baselines.

2 Methodology

2.1 Problem Formulation

At decoding time step t , given the input x and preceding sequences $y_{<t}$, a pretrained auto-regressive LLM θ computes the logit $\mathbf{z}_t \in \mathbb{R}^{|V|}$, where V is the vocabulary, for the t -th token. In the open-domain QA task, a question q serves as the input x , and $\mathbf{z}_t$ relies solely on LM’s parametric knowledge. When both q and the retrieved context c are provided as x , the logit is denoted as $\mathbf{z}_t^c \in \mathbb{R}^{|V|}$.

2.2 Contrastive Decoding

In cases where context cannot be blindly trusted, directly following the context-augmented distribution can increase the risk of being misled. Thus, we adopt the approach of adding the contextual influence, which contrasts with the LLM’s parametric knowledge, to the parametric distribution $\mathbf{z}_t$. With the contrastive decoding objective, $\mathbf{z}_t^c$ and $\mathbf{z}_t$ are ensemble to reflect the influence of external context on LLM’s parametric knowledge at each decoding step t . The probability distribution $P_\theta(Y_t|x, y_{<t})$ is modified by weighted adjustment based on the difference between $\mathbf{z}_t^c$ and $\mathbf{z}_t$, as represented in the following equation.

$$P_\theta(Y_t | x, y_{<t}) = \text{softmax}(\mathbf{z}_t + \alpha (\mathbf{z}_t^c - \mathbf{z}_t)) \quad (1)$$

The contrastive adjustment enables the LLM to integrate external context c into its prediction, leveraging the weight α to control the impact of c on the final probability distribution.

2.3 Adaptive Weight on Contextual Influence

The degree to which contextual influence is incorporated into $\mathbf{z}_t$ needs to be controlled based on the provided context’s informativeness. In practice, however, it is often unknown whether the context is gold or noisy. To address this, we investigate whether the model could adjust accordingly with a simple entropy-based approach.

LLM’s uncertainty is expressed with the entropy $H(Y_t)$ on its probability distribution $P_\theta(Y_t|x, y_{<t})$

(Huang et al., 2023; Kuhn et al., 2023). While $H(Y_t)$ reflects how much uncertainty the model has based on its parametric knowledge under the given question, $H(Y_t^c)$ is influenced by the external knowledge within the retrieved context c . Generally, when the context is added, the entropy decreases. However, if the context is noisy, irrelevant, or provides no information to answer the given question, it may contribute to increased uncertainty instead.

Intuitively, if the retrieved context provides informative cues for answering the question, then $H(Y_t^c)$ is expected to be lowered compared to $H(Y_t)$. Conversely, if the context is non-helpful or even confusing the model prediction, $H(Y_t^c)$ in predicting the next token is likely to be higher. This scenario would be particularly evident when LLM knows the answer with low $H(Y_t)$.

Considering the above scenarios, the motivation behind the adaptive weight α_{ACD} is to assign a relatively smaller weight in cases where the context increases uncertainty by being uninformative or confusing for LLM in answering the given question. Thus, the value of α_{ACD} is set as the proportion of uncertainty contributed by $H(Y_t)$ relative to the total uncertainty when considering both $H(Y_t)$ and $H(Y_t^c)$:

$$\alpha_{ACD} = \frac{H(Y_t)}{H(Y_t) + H(Y_t^c)} \quad (2)$$

Under the condition where $H(Y_t) > H(Y_t^c)$, α_{ACD} value approaches to 1, indicating that when the context c is provided, the uncertainty associated with predicting the next token decreases. Conversely, when $H(Y_t) < H(Y_t^c)$, α_{ACD} value approaches to 0, reflecting minimal influence from c . Note that when $H(Y_t) = H(Y_t^c)$, α_{ACD} becomes 0.5, resulting in an ensemble of two distributions, $\mathbf{z}_t$ and $\mathbf{z}_t^c$, with equal weighting.

With α_{ACD} , the full ACD is obtained, and the vocab v with maximum probability is selected as the next token under the following distribution:

$$\hat{P}_\theta(Y_t | x, y_{<t}) = \text{softmax}(\mathbf{z}_t + \alpha_{ACD} (\mathbf{z}_t^c - \mathbf{z}_t)) \quad (3)$$

Informed by α_{ACD} and contextual contrast, the adjustment process determines the degree to which the model’s parametric knowledge is superseded, thus optimizing the assimilation of contextual information throughout decoding.

Dataset ($\rightarrow$)		TriviaQA			NQ			PopQA		
Model	Method ($\downarrow$)	All	Subset _{Gold}	Subset _{Noisy}	All	Subset _{Gold}	Subset _{Noisy}	All	Subset _{Gold}	Subset _{Noisy}
LLAMA2 7B	Reg _{CLS}	59.00	-	-	25.48	-	-	28.36	-	-
	Reg _{OPn}	60.23	87.40	33.50	31.39	61.31	12.40	38.49	81.21	7.77
	CAD	49.02	73.69	24.75	25.57	51.61	9.05	33.70	72.18	6.03
	MICD _F	60.36	85.72	35.39	29.45	56.10	12.54	35.73	74.25	8.03
	MICD _D	63.23	86.03	40.79	30.36	52.18	16.52	39.01	77.39	11.42
	ACD	64.85	88.01	42.06	32.91	56.60	17.88	41.29	82.77	11.46
LLAMA2 13B	Reg _{CLS}	63.77	-	-	30.80	-	-	32.70	-	-
	Reg _{OPn}	62.81	88.52	37.51	33.35	62.96	14.58	40.03	83.20	8.98
	CAD	52.62	76.78	28.85	27.87	55.96	10.05	35.86	76.38	6.71
	MICD _F	63.53	87.40	40.04	32.63	59.67	15.48	38.16	77.04	10.21
	MICD _D	66.52	87.68	45.69	34.38	57.32	19.83	41.65	79.27	14.60
	ACD	67.37	89.36	45.74	36.12	61.17	20.24	43.35	83.98	14.14

Table 1: EM accuracy of full data (All) and subsets with gold (Subset_{Gold}) and noisy contexts (Subset_{Noisy}).

3 Experimental Results

3.1 Experimental Settings

Datasets and Models We conduct experiments on open-domain QA datasets, TriviaQA (Joshi et al., 2017), Natural Questions (NQ; Kwiatkowski et al., 2019), and PopQA (Mallen et al., 2022) with Wikipedia contexts¹.

We use auto-regressive language models, LLAMA2 (7B & 13B, Touvron et al., 2023), LLAMA3 8B², and MISTRAL 7B (Jiang et al., 2023). Utilizing CONTRIEVER-MSMARCO (Izacard et al., 2022) as a retriever, the top-1 retrieved context is appended to each question.

Evaluation Metric Following Zhao et al. (2024), we use few-shot prompts with 5 examples. We report Exact Match (EM) as an evaluation metric, which verifies whether the generated sequences precisely match one of the candidate answers.

Baselines As fundamental baselines, regular greedy decoding has been employed in open-book (Reg_{OPn}) and closed-book (Reg_{CLS}) settings. We compare our method against existing context-aware contrastive decoding methods, including Context-Aware Decoding (CAD; Shi et al., 2023) and Multi-Input Contrastive Decoding (MICD; Zhao et al., 2024). MICD presents two methods, referred to as MICD_F and MICD_D, which offer fixed and dynamic α , respectively. Similar to our approach, to leverage the burden of hyperparameter search and dependency on fixed α , MICD_D determines α in use by comparing the highest token probabilities with and without given context.

Other implementation details can be found in Appendix B.

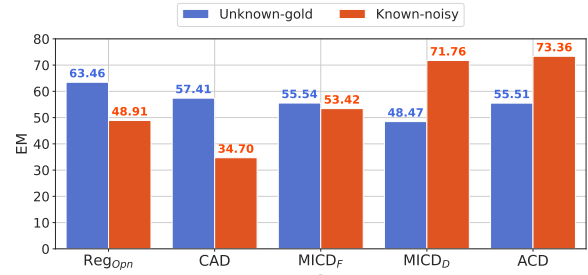


Figure 1: EM accuracy of each method in LLAMA2-7B. EM of three datasets used are averaged for each subset, *Unknown-gold* and *Known-noisy*.

3.2 Main Results

Performance on RAG As shown in Table 1, ACD outperforms the baselines across all datasets and models, especially considering the full test data (All). Similar trends in performance are observed for LLAMA3 8B and MISTRAL 7B, as detailed in Appendix C. When analyzing the performance by dividing the data into two subsets based on whether the retrieved context is gold (Subset_{Gold}) or not (Subset_{Noisy}), ACD achieves either the best or second-best performance. MICD_D demonstrates performance comparable to ACD on Subset_{Noisy}. However, it shows a significant drop on Subset_{Gold}, indicating a tendency to ignore gold context while handling noisy context. It is notable that both CAD and MICD_F exhibit a significant drop in their performance under noisy conditions.

Performance under Parametric Knowledge

We aim to analyze the model’s performance across various aspects, focusing specifically on its parametric knowledge. We estimate whether the model possesses relevant parametric knowledge for a given question based on its accuracy when only the question is provided. We consider two subsets under the following conditions: (1) *Known-noisy*: the

¹Wikipedia dump from Dec. 2018.

²<https://github.com/meta-llama/llama3>

	α	NQ	TriviaQA	PopQA
Max	MICD _D	51.53	59.76	65.49
	ACD	65.78	73.37	74.84
Avg.	MICD _D	54.18	63.78	72.64
	ACD	68.80	72.32	78.90
First	MICD _D	53.92	62.95	68.81
	ACD	73.27	80.45	80.08

Table 2: AUROC between α used in each method and the noisiness of the retrieved context.

model has parametric knowledge and noisy context is retrieved. (2) *Unknown-gold*: the model does not have parametric knowledge and gold context is retrieved.

From Figure 1, we observe that ACD outperforms the baselines in *Known-noisy*. Notably, two approaches with adaptively adjusted weight, ACD and MICD_D, perform well in *Known-noisy*, while other baselines show a relative strength in *Unknown-gold*. However, these baselines also experience significant performance drops in *Known-noisy*, indicating distraction by noisy context despite correctly answering when only the question is provided. In both cases, ACD demonstrates better performance compared to MICD_D, overall showing a tendency towards reliability. The results from other models are in Appendix D.

3.3 Analysis

Correlation between Adaptive Weight and Context Noisiness While other baselines rely on the fixed hyperparameter of weight α , ACD and MICD_D adjust α during the decoding step. It depends not only on the noisiness of the retrieved context but also on whether the model’s parametric knowledge contains an answer to the given question. To exclude cases that are not directly related to the analysis of how weight is adjusted based on context quality, we use the same subsets, *Known-noisy* and *Unknown-gold*.

Adaptive weights α_{ACD} and α_{MICD} are extracted at each decoding step and analyzed across three metrics: maximum, average, and the first within the sequence. As a metric, AUROC between α and the noisiness of the retrieved context is measured. AUROC of each α for LLAMA 2-7B is reported in Table 2, whereas for other models, it can be found in Appendix E. Under every metric and dataset, ACD demonstrates a higher AUROC compared to MICD_D. Aligned with our motivation,

	All	Subset _{Gold}	Subset _{Noisy}
NQ			
ACD	32.91	62.03	18.65
$\alpha = 0.5$	31.91 (-1.00)	58.10 (-3.93)	18.74 (+0.09)
TriviaQA			
ACD	64.85	89.20	43.81
$\alpha = 0.5$	64.32 (-0.53)	88.65 (-0.55)	43.35 (-0.46)
PopQA			
ACD	41.29	84.48	13.60
$\alpha = 0.5$	40.16 (-1.13)	81.93 (-2.55)	13.42 (-0.18)

Table 3: Comparison of EM accuracy between ACD with fixed α as 0.5. Values in parentheses indicate the difference between ACD and the fixed α setting.

when the model is knowledgeable and presented with noisy context, α_{ACD} tends to be lower, emphasizing greater reliance on parametric knowledge. Conversely, when the model lacks knowledge and is provided with gold context, α_{ACD} is adjusted to prioritize reliance on the provided context.

Ablation on α_{ACD} To assess whether α_{ACD} contributes performance improvement, we further compare ACD with a fixed α value of 0.5, where the entropy remains unchanged with and without context. In Table 3, it can be observed that using a fixed α of 0.5 results in degraded performance compared to ACD in most cases except for one case in Subset_{Noisy}. When comparing Subset_{Noisy} and Subset_{Gold}, it is evident that using a fixed α of 0.5 leads to larger decrease in performance in Subset_{Gold}. The overall result suggests that ACD, aside from being based on the output distribution without context, also contributes to performance improvement through adaptive α weighting.

As an additional analysis, we provide a case study of ACD to show how the value of entropies contributes to adjust α_{ACD} and further examine ACD on knowledge flipping task in Appendix F and Appendix G.

4 Conclusion

In this work, we mainly tackle handling noisy contexts in open-domain QA on the RAG framework. This method dynamically adjusts contextual influence during decoding by quantifying uncertainty dropped by the retrieved context. Our results show that ACD improves performances across various dimensions by considering LLM’s parametric knowledge and context noisiness. These findings highlight ACD’s potential to enhance the reliability of retrieval-augmented LLMs.

Limitations

Similar to other contrastive decoding approaches, the inference cost of our approach is higher than the conventional greedy decoding. Specifically, while CAD incurs twice the inference cost and MICD incurs three times the cost, ACD also incurs twice the inference cost of conventional greedy decoding.

Our research is limited the base models and does not encompass chat or instruction-following models trained with reinforcement learning from human feedback (RLHF) or instruction fine-tuning (Ouyang et al., 2022; Chung et al., 2022). These aligned models often generate token distributions that vary significantly based on the presence or absence of contextual instruction or templates. For instance, an instruction-following model might start its generation with "According to the given context ..." when context is provided, while directly generating the answer in absence of context. This alignment with the provided instructions poses another challenge to be tackled when the contrastive decoding approach is utilized.

Our current focus is primarily on short-form QA tasks. Expanding to QA tasks with long-form generation will enable a wider range of applications. Under long-form QA tasks, our approach can be further developed to investigate scenarios where the context is only partially relevant to the question.

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A Related Works

Context-augmented Generation With the objective of closely adhering to the context provided, context-augmented generation strategies have been developed to better utilize the context and produce more factual and context-relevant responses at the inference time (Zhou et al., 2023; He et al., 2024). Especially, approaches integrated with contrastive decoding (Li et al., 2023; Malkin et al., 2022; Liu et al., 2021) yield significant enhancement by amplifying the influence of the given context at decoding step (Shi et al., 2023; Zhao et al., 2024). These context-aware contrastive decoding methods to generate responses faithful to the given context show effective performance in summarization (See et al., 2017; Narayan et al., 2018), knowledge flipping (Longpre et al., 2022), and question answering with gold-standard contexts.

Answer the following questions:

<few-shots>

Question: <question>

Answer:

Table 4: Template used in closed-book generation.

Answer the following questions:

<few-shots>

Context: <context>

Question: <question>

Answer:

Table 5: Template used in open-book generation.

Robustness in RAG Framework While retrieval-augmented generation enables LLMs to become factual and reliable with the retrieved external knowledge, there are still concerns about incorrectly retrieved irrelevant contexts (Yoran et al., 2024). To address hallucination errors posed by irrelevant contexts, some researchers take an approach to train LLMs that can adaptively retrieve relevant context (Asai et al., 2023b; Wang et al., 2024). Another approach aims to selectively use retrieved contexts after assessing their truthfulness or relevance through context verification with prompting strategies or training untruthful context detectors (Yu et al., 2024; Zhang et al., 2024). These approaches highlight the ongoing efforts to advance the robustness and accuracy of LLMs in multiple directions to manage potentially misleading information.

B Implementation Details

B.1 Instructions

The templates we use throughout the experiment are in Table 4 and Table 5. The template used in open-book generation (Table 5) is applied to get context-augmented distribution $\mathbf{z}_t^c$. Also, to obtain $\mathbf{z}_t$, the template in Table 4 is used.

B.2 Datasets

For NQ and TriviaQA, general world knowledge is required to answer the given question. In PopQA, tackling long-tailed information, less popular factual knowledge is asked. For NQ and TriviaQA,

	R@1	R@5	R@10	R@20	R@100
NQ	38.81	65.65	73.91	79.56	88.01
TriviaQA	49.60	71.32	76.72	80.39	85.71
PopQA	41.83	61.54	68.63	74.55	83.95

Table 6: Recall@100 performance for CONTRIEVER-MSMARCO

few-shot examples are adopted from train data. For PopQA, we randomly sample 5 examples with different relationship types for sample diversity. The number of test data in used is 3,610 for NQ, 11,313 for TriviaQA, and 14,262 for PopQA.

B.3 Baselines

CAD introduces a context-aware contrastive decoding approach that employs a contrastive output distribution to accentuate discrepancies in model predictions with and without context. This method effectively overrides model priors conflicting with provided context, offering significant performance enhancements in tasks requiring resolution of knowledge conflicts. MICD further enhances context grounded generation by integrating contrastive decoding with adversarial irrelevant passages.

From a computational time perspective, MICD requires three times more than conventional greedy decoding, while CAD and ACD require twice as much.

MICD proposes two usage directions, referred to as $MICD_F$ and $MICD_D$, which offer fixed and dynamic α , respectively. In $MICD_D$, α is assigned as the maximum token probability with context ($\max P_{wc}$) if $\max P_{wc}$ exceeds the maximum token probability without context ($\max P_{woc}$); otherwise, it is calculated as $1 - \max P_{woc}$.

Throughout the experiments, fixed value of α is set to the value used in Zhao et al. (2024), 0.5 and 1.0 for CAD and $MICD_F$, respectively.

B.4 Retriever Performance

To assess performance in the RAG framework, the top-1 context from top-100 contexts retrieved by CONTRIEVER-MSMARCO (Izacard et al., 2022) is utilized. Recall@100 is reported for each dataset in Table 6.

C Main Results on Llama3 and Mistral

Main results on LLAMA3 8B and MISTRAL 7B are reported in Table 7.

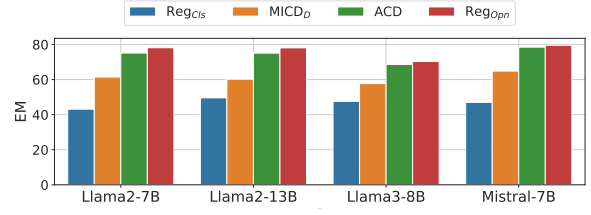


Figure 2: EM accuracy on NQ-swap with contexts replacing the gold answer with a random entity span.

D Full Results on Known-noisy and Unknown-gold

For *Known-noisy* and *Unknown-gold*, the exact values of EM accuracy on each case are reported in Table 8 and Table 9, respectively.

E AUROC between α and Context Noisiness

AUROC of ACD and $MICD_D$ for three models not reported in Table 1 is reported in Table 10.

F Case Study on α_{ACD}

We conduct the case study on α_{ACD} , examining its value in cases of *Known-noisy* and *Unknown-gold*. Table 11 shows the generations from LLAMA2 7B and how the values of entropy from closed-book generation (Reg_{CIS}) and open-book generation (Reg_{Opn}) affect α_{ACD} at the first decoding time step.

In case of *Known-noisy*, when the model generates the answer correctly even without the given context, the retrieved noisy context yields relatively higher entropy, resulting in α_{ACD} value of 0.3483.

Conversely, in the case of *Unknown-gold*, the model’s generated answer is incorrect, aligning with a relatively high entropy value of 6.6748. In this scenario, the retrieved gold context guides the model to correctly answer the question, which is reflected in a relatively lower entropy value of 1.5628. Thus, the value of α_{ACD} , adjusted with these entropy values, yields a relatively higher weight on the context at 0.8103.

G Knowledge Conflict

Knowledge Conflict In case where the provided context is different from what the model knows but relevant to the question, we verify whether the two decoding methods with dynamic weight, ACD and $MICD_D$, can generate responses without considering it as a noisy context with NQ-Swap dataset

(Longpre et al., 2022).

NQ-swap (Longpre et al., 2022) is constructed by replacing the answer entity span in the original gold context of NQ questions with a random entity, to verify if the knowledge is successfully flipped. We use the questions and swapped contexts provided in Hong et al. (2024) with 3650 samples excluding 5 examples used as few-shot prompts and samples with the context in a tabular format due to the limited context length. In the case of NQ-SWAP, each data point has a given context. Since it is a task that does not use a retriever, for MICD, we use the fixed negative context taken from the MICD as an adversarial context. MICD reports that the performance difference between fixed negative and the most-distant context is negligible.

Figure 2 illustrates that ACD consistently exceeds the performance of MICD_D across all models and achieves results comparable to open-book regular decoding. The results indicate that the ACD’s approach remains effective even in settings where the context is relevant to the question but contradicts the parametric knowledge.

Datasets ($\rightarrow$)	TriviaQA			NQ			PopQA		
Methods ($\downarrow$)	All	Subset _{Gold}	Subset _{Noisy}	All	Subset _{Gold}	Subset _{Noisy}	All	Subset _{Gold}	Subset _{Noisy}
LLAMA3 8B									
Reg _{Cls}	61.67	-	-	28.34	-	-	32.65	-	-
Reg _{Opn}	61.27	<u>86.94</u>	36.02	<u>33.30</u>	63.10	14.40	39.73	<u>82.95</u>	8.64
CAD	49.70	<u>72.45</u>	27.31	29.17	58.39	10.64	35.86	<u>76.82</u>	6.40
MICD _F	61.01	85.40	37.00	27.62	51.89	12.22	37.99	<u>77.12</u>	9.85
MICD _D	<u>64.01</u>	86.08	<u>42.28</u>	30.72	53.96	<u>15.98</u>	<u>41.35</u>	79.32	14.04
ACD	66.32	89.20	43.81	35.48	<u>62.03</u>	18.65	43.25	84.48	<u>13.60</u>
MISTRAL 7B									
Reg _{Cls}	63.72	-	-	29.64	-	-	29.04	-	-
Reg _{Opn}	60.45	86.85	34.48	32.55	64.67	12.18	38.28	<u>81.26</u>	7.36
CAD	44.69	66.89	22.85	24.10	52.25	6.25	33.93	<u>73.95</u>	5.15
MICD _F	63.33	88.43	38.62	31.80	61.10	13.22	36.58	76.00	8.23
MICD _D	<u>66.97</u>	<u>89.24</u>	<u>45.05</u>	<u>33.24</u>	<u>57.89</u>	<u>17.61</u>	<u>39.87</u>	<u>78.46</u>	12.11
ACD	67.82	90.16	45.83	35.37	<u>62.17</u>	18.38	41.47	82.90	<u>11.68</u>

Table 7: EM accuracy of full data (All) and subsets with gold context (Subset_{Gold}) and with noisy context (Subset_{Noisy}).

	NQ	TriviaQA	PopQA
LLAMA2-7B			
Reg _{Opn}	45.13	68.12	33.47
CAD	29.22	48.91	25.97
MICD _F	51.07	72.37	36.81
MICD _D	<u>72.92</u>	<u>86.33</u>	<u>56.04</u>
ACD	76.72	88.79	54.58
LLAMA2-13B			
Reg _{Opn}	47.18	69.77	32.53
CAD	32.04	52.48	22.66
MICD _F	54.17	75.05	38.55
MICD _D	76.31	<u>88.24</u>	59.38
ACD	<u>75.15</u>	88.78	<u>56.11</u>
LLAMA3-8B			
Reg _{Opn}	46.20	68.50	33.39
CAD	32.91	50.51	23.00
MICD _F	43.25	70.67	39.07
MICD _D	<u>61.18</u>	<u>83.70</u>	59.80
ACD	64.14	86.59	<u>56.87</u>
MISTRAL-7B			
Reg _{Opn}	41.04	64.57	31.03
CAD	19.17	42.63	20.80
MICD _F	48.12	71.99	36.48
MICD _D	<u>69.58</u>	<u>86.84</u>	57.14
ACD	70.62	89.36	<u>53.55</u>

Table 8: EM accuracy of *Known-noisy* case.

	NQ	TriviaQA	PopQA
LLAMA2-7B			
Reg _{Opn}	47.78	68.18	74.42
CAD	<u>43.90</u>	<u>62.22</u>	66.12
MICD _F	40.47	61.51	64.63
MICD _D	29.82	50.43	65.17
ACD	36.03	57.10	<u>73.41</u>
LLAMA2-13B			
Reg _{Opn}	46.52	65.09	75.04
CAD	<u>45.77</u>	61.07	69.43
MICD _F	41.79	<u>62.03</u>	65.85
MICD _D	30.72	47.77	64.70
ACD	36.19	53.98	<u>72.38</u>
LLAMA3-8B			
Reg _{Opn}	48.12	68.47	74.10
CAD	48.00	60.52	70.41
MICD _F	38.15	<u>61.40</u>	65.55
MICD _D	33.33	48.45	64.81
ACD	41.67	61.24	<u>72.52</u>
MISTRAL-7B			
Reg _{Opn}	49.57	64.82	73.09
CAD	<u>45.38</u>	56.02	67.81
MICD _F	43.28	<u>63.59</u>	66.58
MICD _D	32.06	54.97	66.37
ACD	37.73	57.70	<u>73.03</u>

Table 9: EM accuracy of *Unknown-gold* case.

	α	NQ	TriviaQA	PopQA
LLAMA2 7B				
Max	MICD _D	52.77	60.09	61.84
	ACD	69.24	75.31	74.12
Avg.	MICD _D	57.86	62.00	71.79
	ACD	71.61	73.41	77.92
First	MICD _D	54.80	46.13	68.44
	ACD	73.07	77.96	80.51
LLAMA3 8B				
Max	MICD _D	50.75	52.59	63.72
	ACD	63.12	57.82	75.00
Avg.	MICD _D	51.80	52.83	67.99
	ACD	64.08	59.67	75.90
First	MICD _D	45.70	39.07	69.21
	ACD	67.48	75.45	80.31
MISTRAL 7B				
Max	MICD _D	56.98	64.95	61.93
	ACD	71.27	77.46	74.11
Avg.	MICD _D	63.66	69.27	73.82
	ACD	76.02	78.20	79.08
First	MICD _D	56.84	68.98	71.73
	ACD	75.75	84.11	82.07

Table 10: AUROC between α used in each method and the noisiness of the retrieved context. The best AUROC is in bold.

Sample		Reg _{Cls}		Reg _{Opn}		ACD	
Case		Generation	$H(Y_t)$	Generation	$H(Y_t^c)$	Generation	α_{ACD}
<i>Known-noisy</i>	Question: who does the voice of nala in the lion king? Gold answer: Moira Kelly	Moira Kelly	2.9160	Whoopi Goldberg	5.4562	Moira Kelly	0.3483
<i>Unknown-gold</i>	Question: who was the actor that played ben stone on law and order? Gold answer: Michael Moriarty	Michael Tucker	6.6748	Michael Moriarty	1.5628	Michael Moriarty	0.8103

Table 11: Case study on the value of α_{ACD} for *Known-noisy* and *Unknown-gold* cases. Each value of entropy without context ($H(Y_t)$), entropy with context ($H(Y_t^c)$), and α_{ACD} is extracted at the first decoding step ($t = 0$). LLAMA2 7B is used.